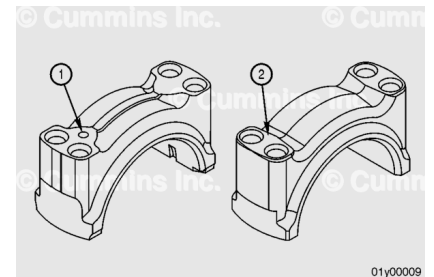


Trainings ▾

General Information

⚠ CAUTION ⚠

The fracture-split connecting rod requires a different large end rod bearing than the saw cut connecting rod. The bearings are specific to the connecting rod design and are not backward compatible. The oil drilling in the connecting rod and upper bearing shell will not be aligned if the wrong bearing is used. If the bearings are mismatched with the rod design, severe engine damage could occur.



⚠ CAUTION ⚠

To prevent damage to the fracture-split connecting rod, do not set the connecting rod or rod cap on the fracture-split connection. This can cause polishing and damage to the mating surface.

There are two styles of connecting rod assemblies on the ISX15 and QSX15 engines.

Newer model engines use a fracture-split (1) connection between the connecting rod and connecting rod cap.

Older model engines use a saw cut (2) connection between the connecting rod and connecting rod cap.

It is important to understand if the rod has a fracture-split or a saw cut connection before continuing with this procedure. For identification, the fracture-split connecting rod will have a visible hole drilled in the cap.

Note : Saw cut connecting rod assemblies can be mixed with fracture-split connecting rod assemblies within the same engine.

Connecting rods with a fracture-split surface **must** be treated with additional caution. The two pieces of the connecting rod can **not** be rubbed together. This will damage the mating surfaces. Use care to **not** drop either piece of the connecting

rod. Fracture-split connecting rods **must only** be handled if the two pieces of the connecting rod are tightened to the correct specification, or completely separated.

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

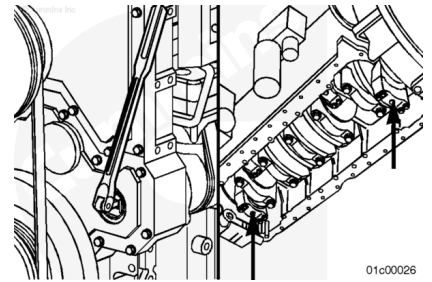


ck800wa

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html>)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-046-tr.html>)

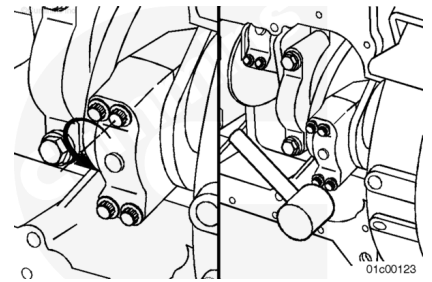
Remove

Rotate the crankshaft so two of the connecting rods are at bottom dead center.



⚠ CAUTION ⚠

Do not damage the fracture-split surface on the connecting rod or connecting rod cap while the connecting rod caps are removed. If the fracture-split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the connecting rod.



Do **not** remove the capscrews at this time.

Loosen the connecting rod capscrews.

Hit the connecting rod capscrews with a rubber hammer to loosen the rod caps from the dowels.

Remove the capscrews and the rod cap. Note the corresponding rod and rod caps, as well as the orientation of the rod cap.

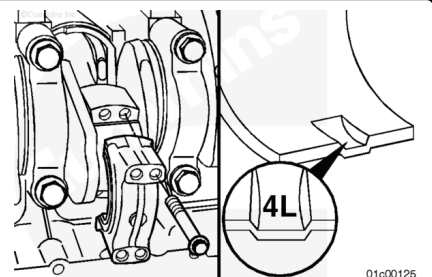
Use care **not** to damage the fracture-split mating surface.

Do **not** set the rod cap with the fracture-split side down.

Note : Both the rod cap and the connecting rod are serialized on one face to make sure of proper orientation.

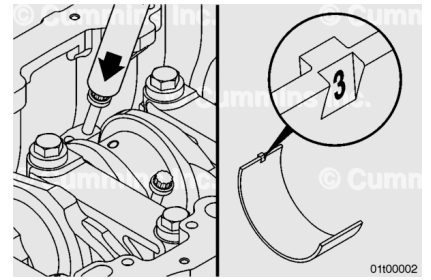
Connecting rod shell bearing upper or lower location is denoted by UPPER or UPR (for connecting rod) and LOWER or LWR (for connecting rod cap) stamped into the back of the bearing shells. The upper connecting rod shell bearing can also be identified by the oil port in the bearing.

Remove the bearing shell from the connecting rod cap. Mark the cylinder number on the flat surface of the bearing tangs.



⚠ CAUTION ⚠

Use caution to avoid damaging the crankshaft journal while pushing up the connecting rod. Only push on the head of the capscrew. Do not push against the fracture-split face of the connecting rod.



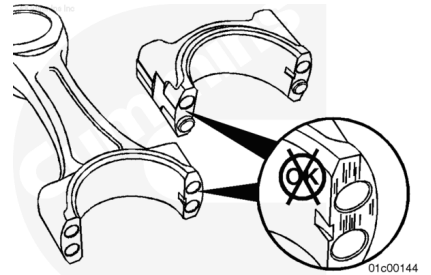
Install two guide pins, Part Number 3163097, in the connecting rod.

Push the rod up far enough to allow the upper bearing shell to be removed. Use a T-handle piston pusher.

Remove the bearing shell. Mark the cylinder number and the letter "U" on the flat surface of the bearing tangs.

⚠ CAUTION ⚠

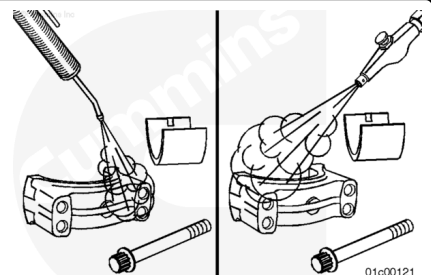
Prevent damage to the fracture-split connecting rod. Do not set the connecting rod or rod cap on the fracture-split connection. This can cause polishing and damage to the mating surface.



Clean and Inspect for Reuse

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

Use mineral spirits to clean the parts.

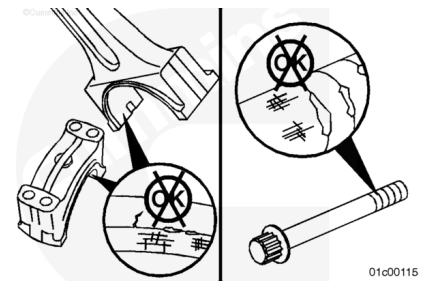
Dry with compressed air.

Inspect the rod caps, connecting rod bearing saddles, and capscrews for nicks, cracks, burrs, scratches, or fretting.

Saw Cut Connecting Rods

- Inspect the connecting rod and connecting rod cap for fretting damage on the mating surfaces.
- The connecting rod and connecting rod cap **must** be replaced as an assembly if fretting damage is visible on either piece.
- Polishing can be observed on the thrust face of the connecting rod or rod cap which is normal. Use the following procedure for inspection criteria. Refer to Procedure 001-007 in Section 1.

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Fracture Split Connecting Rods

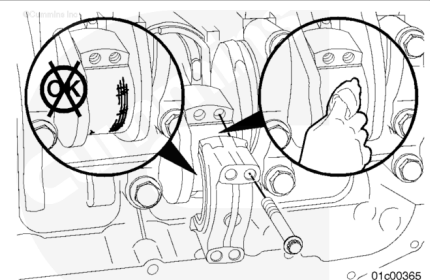
- Polished spots on either the connecting rod or connecting rod cap mating surfaces indicate fretting. The rod **must** be replaced. The connecting rod and connecting rod cap **must** be replaced as an assembly if fretting damage is visible on either piece.
- Dark areas on the face indicate carbonized oil deposits. Connecting rods with dark areas are reusable if there is no fretting present.
- The fracture joint can produce debris at each separation. Use solvent and a nylon bristle brush to clean the contact surfaces between the connecting rod and connecting rod cap.
- Polishing can be observed on the thrust face of the connecting rod or rod cap, which is normal. Use the following procedure for inspection criteria. Refer to Procedure 001-007 in Section 1.

(/qs3/pubsys2/xml/en/procedures/10/10-001-007-tr.html)

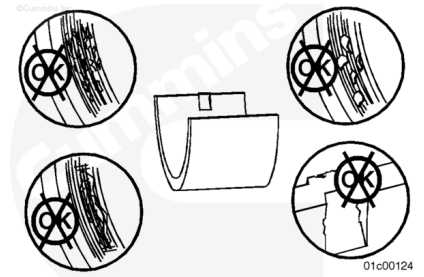
Inspect the connecting rod journal of the crankshaft for scratches or nicks.

Use a crocus cloth to remove nicks and scratches.

Clean off any dust or debris released while using the crocus cloth.



Fracture-split connecting rods require a different large end rod bearing from the saw cut connecting rod. An engine can have a mixture of fracture and saw cut connecting rods. It is critical to install the correct rod bearings in the correct connecting rod design. Failure to do so could result in severe engine damage.

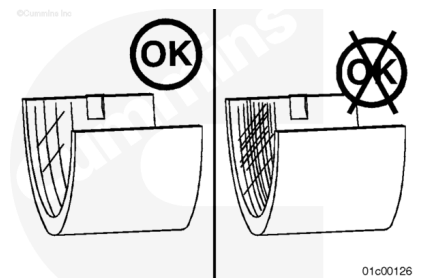


Inspect the bearings for damage.

Replace any bearings with the following damage:

- Pitting
- Flaking
- Corrosion
- Lock tang damage
- Scratches (deep enough to be felt with a fingernail).

Normal bearing wear produces a smooth finish that will wear into the lining. See Parts Reuse Guidelines, Bulletin 3810303.



For more detailed information of bearing damage, see Analysis and Prevention of Bearing Failures, Bulletin 3810387.

Inspect the bearing seating surface for nicks or burrs.

If nicks or burrs can **not** be removed with a fine crocus cloth, the bearings **must** be replaced.

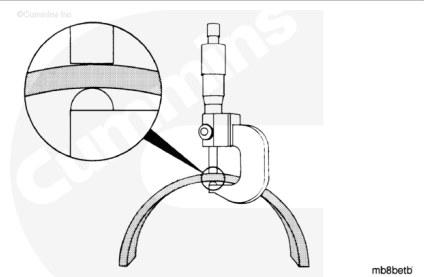
Clean off any dust or debris released while using the crocus cloth.



Measure the rod bearing shell thickness with an outside micrometer that has a ball tip.

Discard a bearing shell if its thickness is below the minimum specification.

Note : The fracture-split style connecting rod uses a thicker bearing than the machined style connecting rod.



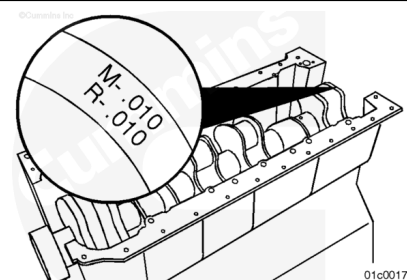
Standard Connecting Rod Bearing Thickness

mm		in
2.366	MIN	0.093
2.378	MAX	0.094

Standard Size, Fracture-Split Style Connecting Rod Bearing Thickness

mm		in
2.776	MIN	0.109
2.788	MAX	0.110

Crankshafts ground undersize in the connecting rod or the main bearing journals are marked on the front counterweight. If the crankshaft is marked, check the bearing shell part number to make sure the correct bearing size is used.

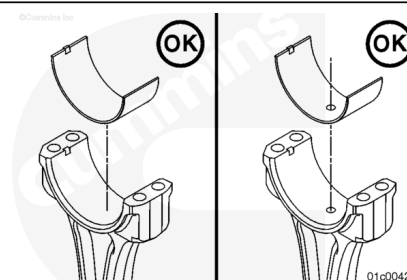


Install

⚠ CAUTION ⚠

Connecting rod bearings must match the connecting rod design. Drilled rods must use drilled rod bearings. Non-drilled rods must use non-drilled bearings. Mismatches will cause engine damage. Also, fracture-split connecting rods must be fitted with fracture-split connecting rod bearings.

Saw cut connecting rods must be fitted with saw cut connecting rod bearings. Mismatches can cause severe engine damage.



⚠ CAUTION ⚠

The fracture-split connecting rod requires a different large end rod bearing from the saw cut connecting rod. An engine can have a mixture of fracture-split and saw cut connecting

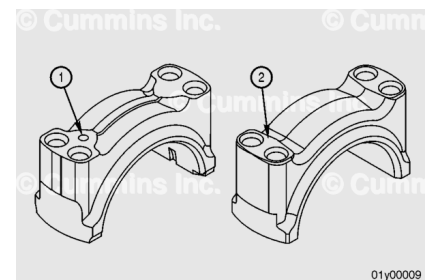
rods. It is critical to install the correct rod bearings in the correct connecting rod design. Failure to do so could result in severe engine damage.

Connecting rod design **must** be matched with the proper connecting rod bearing.

Drilled connecting rods **must** be fitted with drilled connecting rod bearings.

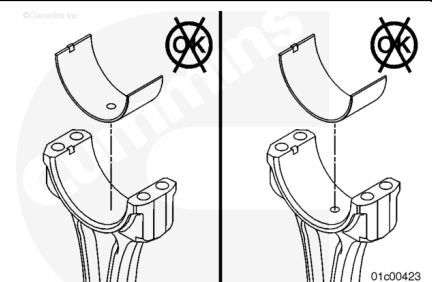
Non-drilled connecting rods **must** be fitted with non-drilled connecting rod bearings.

Fracture-split connecting rods (1) can be mixed in the same engine as saw cut connecting rods (2). It is critical to understand which connecting rods you are working with to make the appropriate bearing shell selection.



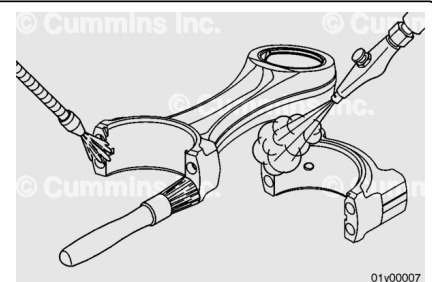
Drilled and non-drilled connecting rods can be mixed in the same engine as long as the two piece piston is used.

Engines using single piece pistons can **not** use non-drilled connecting rods.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

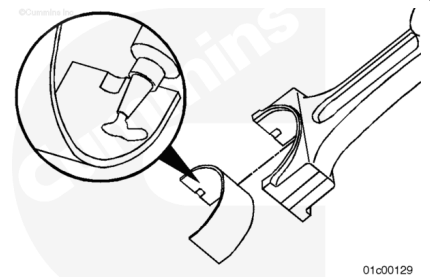
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Fracture-Split Connecting Rods

- The fracture joint can produce debris at each separation. It is important to clean before assembly. Use solvent and a nylon bristle brush to clean the fracture contact surface between the connecting rod and connecting rod cap as well as the bearing saddle area. Dry the contact surface with compressed air.

⚠ CAUTION ⚠

The connecting rod and bearing shell mating surfaces must be clean and dry when the bearing shells are installed. Used bearings must be installed in their original location or engine damage can occur.

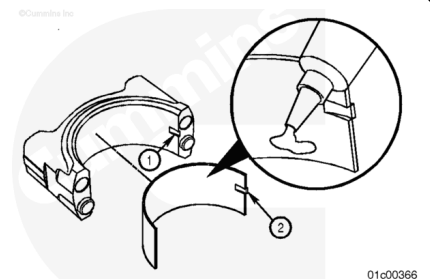


Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the crankshaft journal mating surface of the upper bearing shell.

Install the upper bearing shell into the connecting rod with the tang of the bearing in the slot of the rod.

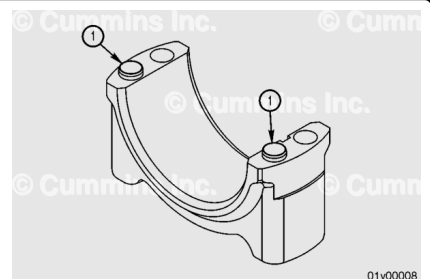
Install the bearing shell into the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

Use clean Lubriplate™ 105 multi-purpose lubricant, or equivalent, to lubricate the bearing shell to crankshaft journal mating surface and connecting rod capscrews.



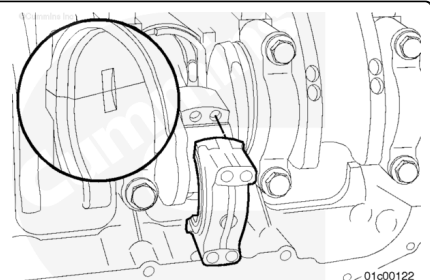
Saw Cut Connecting Rods

- Verify both connecting rod dowel rings are installed in either the connecting rod or connecting rod cap.



⚠ CAUTION ⚠

The rod cap alpha characters must match the alpha characters on the connecting rod and must be installed with the characters aligned to prevent damage to the connecting rods and crankshaft. The locking tang of the connecting rod



cap must face toward the intake side of the cylinder block.
Rod and cap mating surfaces must be clean or engine damage can occur.

Install the cap onto the connecting rod.

Fracture-Split Connecting Rods

- Make sure the rod cap is installed **only** on the connecting rod from which it was removed and the rod cap is properly oriented when installing. Both the rod cap and connecting rod are serialized on one face to make sure of proper orientation. If the connecting rod cap is installed either on the wrong rod or in the wrong orientation and the surfaces are drawn together with the capscrews, the rod **must** be replaced.

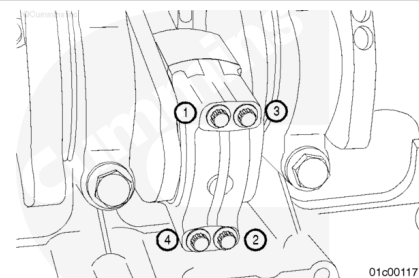
Note : Minimize the number of joint interactions between the connecting rod and rod cap. If the fracture-split mating surface becomes damaged or worn, the rod cap will not fit to the connecting rod as designed and can cause engine damage.

Lubricate the threads of the capscrews with clean engine oil.

Thread the capscrews into the connecting rod. Hand-tighten.

The procedure for tightening used capscrews is different than the procedure for tightening new capscrews. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)

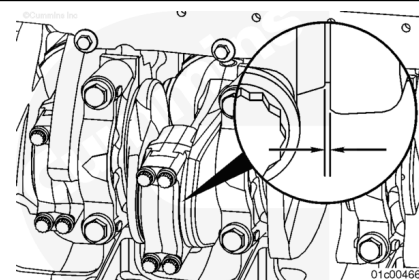


Measure the connecting rod side clearance. Refer to Procedure 001-054 in Section 1.

(/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)

Note : The connecting rod **must** move freely from side-to-side on the crankshaft journal. If the rod does **not** move freely, remove the rod cap and make sure the bearing shells are the correct size. Check for dirt or damage on the crankshaft and bearing shells.

Repeat the above steps to install the remaining bearing shells and connecting rod caps.



Finishing Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



⚠ CAUTION ⚠

The lubricating oil system must be primed before operating the engine after any internal engine repairs to avoid internal component damage.

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-046-tr.html>).
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1 (</qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html>).
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-035.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html>).
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html>).
- Operate the engine until it reaches a temperature of 82°C [180°F]. Check for lubricating oil leaks.

